RAJALAKSHMI ENGINEERING COLLEGE

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T2

Started on	Friday, 28 August 2026, 2:19 PM
State	Finished
Completed on	Friday, 28 August 2026, 2:27 PM
Time taken	8 mins 24 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n, x, i, floor = -1;
    scanf("%d", &n);

    int a[n];
    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    scanf("%d", &x);

    for (i = 0; i < n; i++) {
        if (a[i] <= x)
            floor = a[i];
        else
            break;
    }

    printf("%d", floor);

    return 0;
}
```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔

	Input	Expected	Got	
✔	5 10 22 85 108 129 100	85	85	✔
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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